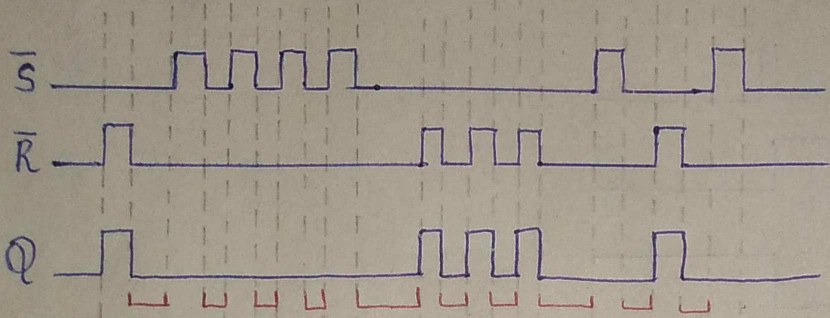
Digital Logic Design

Assignment-3

Name :- Sana Munir Alam
Roll#:- 24K-0573
Class:- BSCS-2E

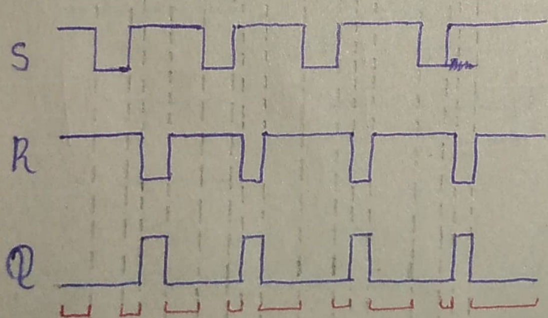
Q1) If the waveforms in Fig-1 are applied to an active-low S-R Latch, draw the resulting Q output waveform in relation to the inputs. Assume that Q starts Low.



[Active Low SR-Latch]

The red highlighted segments indicates invalid state as $\overline{S}$-$\overline{R}$ are both 0.

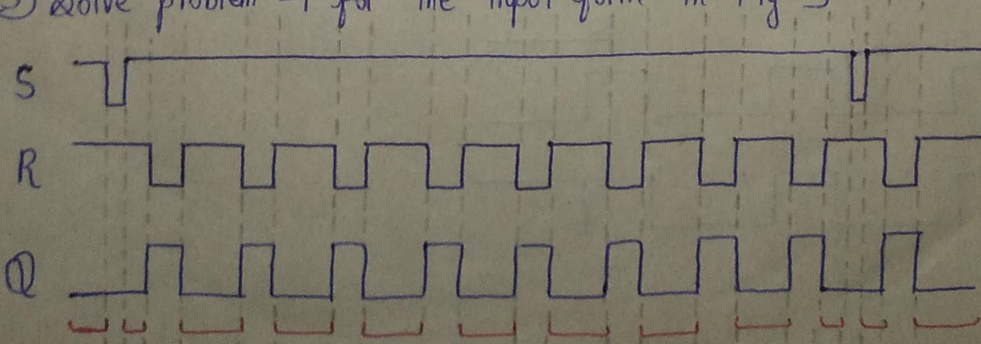
Q2) Solve problem-1 for the input waveform in Fig-2 applied to an active-high SR Latch



[Active-High S-R Latch]

The red highlighted segments indicates invalid state as S-R are both 1.

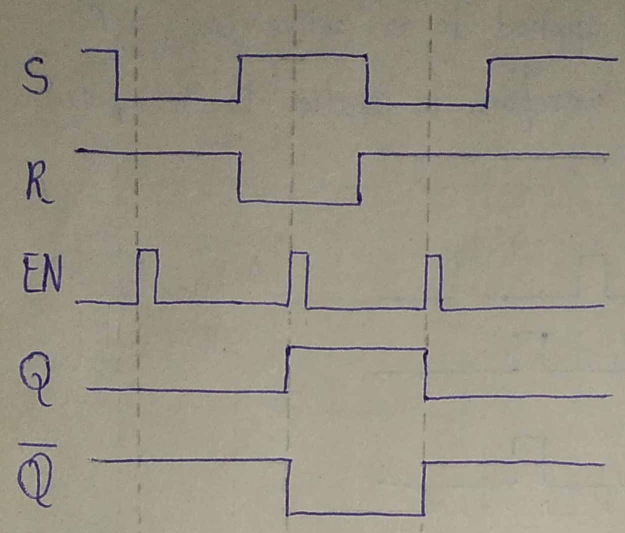
Q3) Solve problem-1 for the input form in Fig-3



[Active High S-R Latch]

The red highlighted segments indicates invalid state as S-R are both 1.

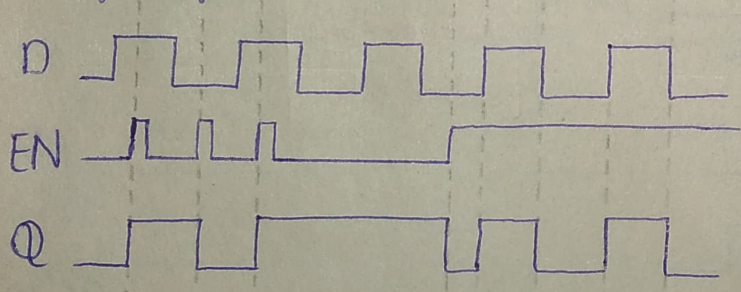
Q4) For a gated S-R latch, determine the Q and Q' output for the inputs in Fig-4. Show them in proper relation to the enabled ~~i/p~~ input. Assume that A starts LOW.



[Active High Gated S-R Latch]

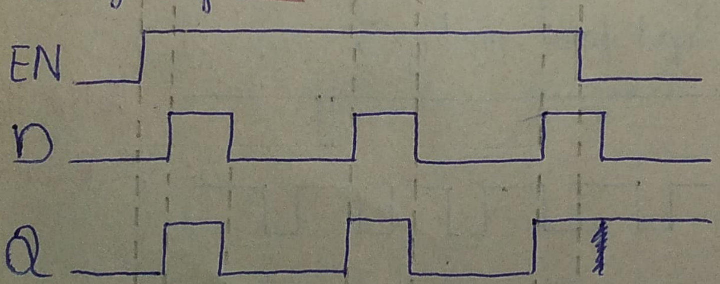
Q5) Determine the output of the gated D latch for the inputs in fig-5 and Fig 6

Solving Figure 5 first :-



[Gated D latch]

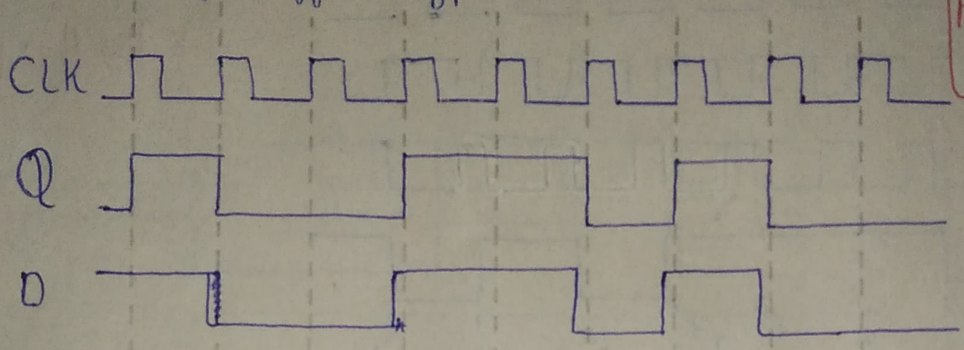
Solving Figure 6



[Gated D latch].

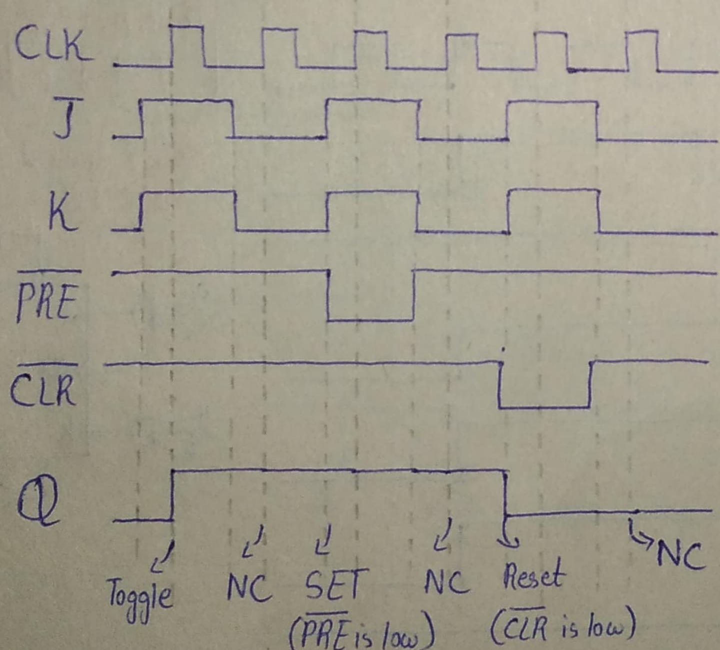
Q6) The Q output of an edge-triggered D flip flop is shown in relatio to the clock signal in Fig-7. Determine the input waveform on the D input that is required to produce this output if the flip-flop is a positive edge-triggered type.

[Positive Triggered D flip flop]

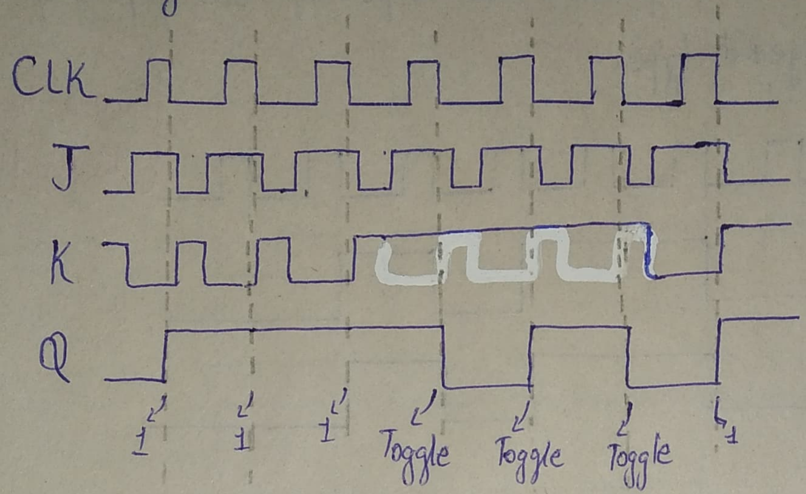


Q7) Determine the Q waveforms relative to the clock if the signal shown in fig-8 are applied to the input of the J-K flip-flop. Assume that Q initially LOW.

[J-K flip flop]



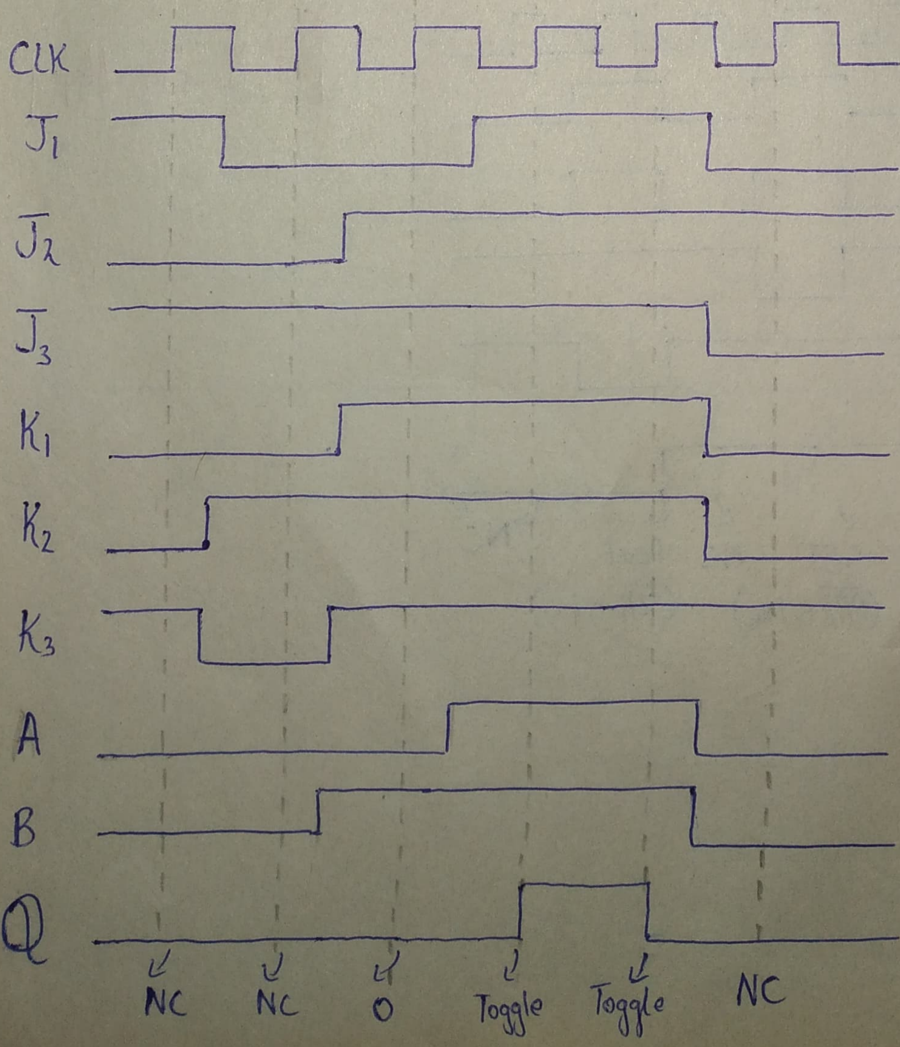
Q8) For negative edge-triggered J-K flip flop with the input in Fig-9 develop the Q output waveform relative to the clock. Assume that Q is initially LOW.



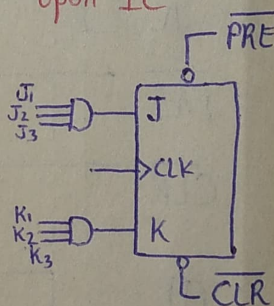
[Negative-edge J-K flip flop]

Q9) For the circuit in Figure-10, complete the timing diagram in Fig-1 by showing the Q output (which is initially LOW). Assume PRE and CLR remain High.

[Positive edge J-K flip flop]



solved based upon IC



Note :-

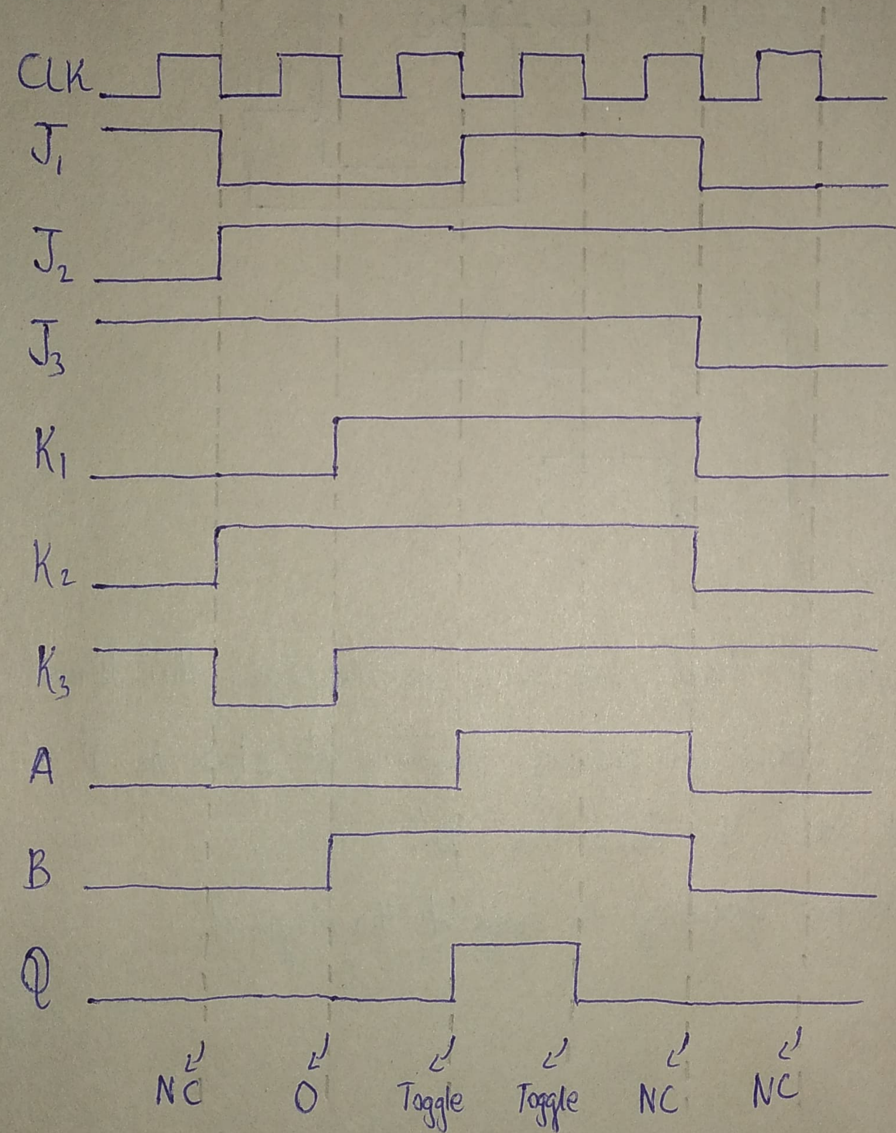
Let $A = J_1 \cdot J_2 \cdot J_3$

$B = K_1 \cdot K_2 \cdot K_3$

Solving Q9 again but this time based upon the negative edge j-k flip flop made in assignment.

- The first solution was based upon IC. 1
- The second solution was based upon waveform.

Solution 2:

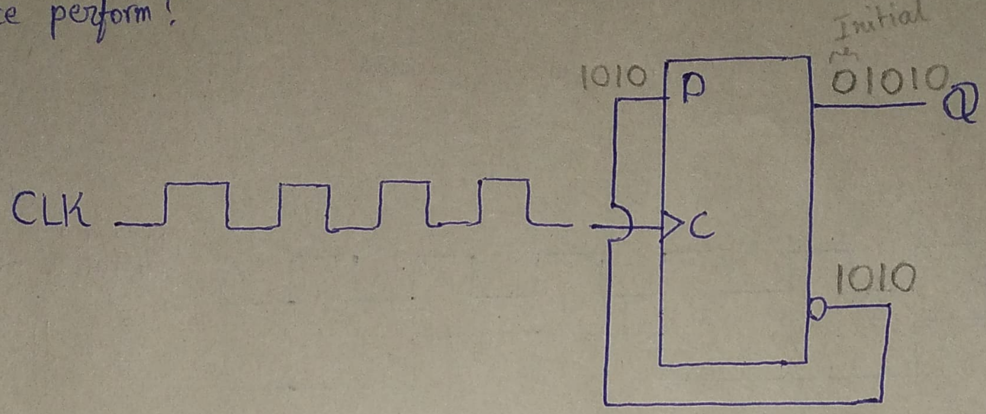


Negative-edge J-K flip flop

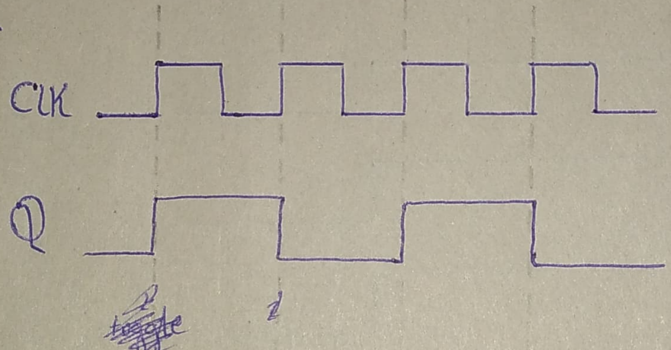
Let $A = J_1 \cdot J_2 \cdot J_3$

$B = K_1 \cdot K_2 \cdot K_3$

Q10) A D flip-flop is connected as shown in fig-12. Determine the Q output in relation to the clock. What specific function does this device perform?



Sol



The flip flop circuit toggles at every positive edge, this kind of device can be used for reducing frequency of clock as $\overline{Q}$ is feeding back onto the data line 'D'.

Therefore, the output frequency is half of the original.